

CRT

1. Difference between event-driven and console-based (in Eclipse):

- Console-based apps run sequentially in the Eclipse Console. Input and output happen through text, and the program controls the flow.
- Event-driven apps (like GUI apps in Swing or JavaFX) wait for events, such as button clicks or mouse movements, and execute code only when an event occurs.

2. Code execution in event-driven app:

- Eclipse runs the program, but the GUI waits for user events. When an event happens (like clicking a JButton), the corresponding event handler method executes. The program doesn't run top-to-bottom like a console app.

3. Can components be added directly to a frame?

- In Java Swing (using Eclipse), components are usually added to the content pane of a JFrame. Adding directly to the frame is allowed in newer Java versions, but using the content pane is best practice.

4. Can a label respond to events?

- Normally, JLabels just display text or images and don't generate events. However, you can add a mouse listener to a JLabel to detect clicks or hover events.

5. Why run GUI from the event-dispatching thread?

- The event-dispatching thread (EDT) ensures that all GUI updates are handled safely. Running GUI code outside the EDT can cause inconsistent behavior, race conditions, or

crashes.

6. Difference between label and button:

- JLabel: displays text or images, not interactive.
- JButton: clickable, triggers events, and can perform actions when pressed.

8. Three ways to control layout in a content pane:

1. Use layout managers like BorderLayout, FlowLayout, or GridLayout.
2. Set absolute positions manually with setBounds().
3. Use nested panels with different layouts to organize components.

13. Numeric data from a text field:

- Data typed in a JTextField is a String. You must convert it to a numeric type using Integer.parseInt() or Double.parseDouble() before performing calculations.
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14. Step by step:

- num2.doubleValue() → 3.0
- Double.valueOf(num3).doubleValue() → 5.0
- num1 = 3.0 + 5.0 → 8.0

15. Better component for selecting from 6 names:

- JComboBox is better than a JTextField because it provides a predefined list, prevents typing errors, and is easier for the user to select from.